

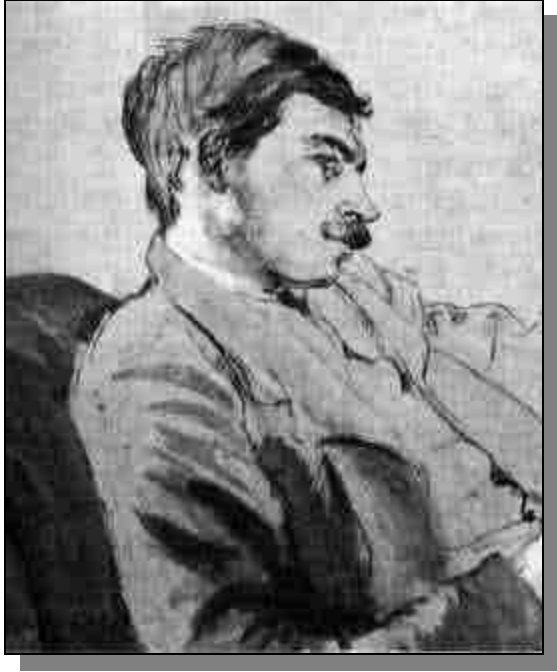
IS-LM Model



The Great Depression caused many economists to question the validity of classical economic theory (from Chapters 3-6). They believed they needed a new model to explain such a pervasive economic downturn and to suggest that government policies might ease some of the economic hardship that society was experiencing.

In 1936, **John Maynard Keynes** wrote *The General Theory of Employment, Interest, and Money*. In it, he proposed a new way to analyze the economy, which he presented as an alternative to the classical theory.

Keynes proposed that low **aggregate demand** is responsible for the low income and high unemployment that characterize economic downturns. He criticized the notion that aggregate supply alone determines national income.

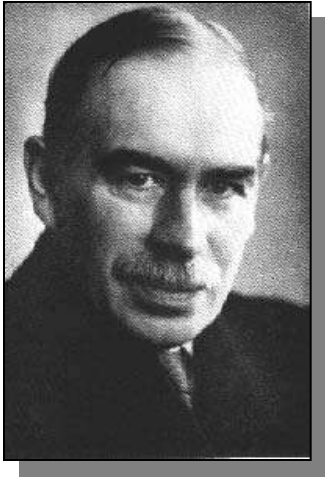


The Keynesian Model

In 2008 and 2009, as the United States and Europe descended into a recession, the **Keynesian theory of the business cycle** was often in the news.

Policymakers around the world debated how best to increase aggregate demand with both **monetary** and **fiscal policy**.

Background on the Model



“**Keynesian**” means different things to different people. It’s useful to think of the basic textbook Keynesian model as an elaboration and extension of the “**classical theory.**”

Its variable velocity of money and “sticky” prices reflects Keynes’s belief that the Classical model’s shortcomings arose from its overly-strict assumptions of **constant velocity** and **highly flexible wages and prices.**

The model of **aggregate demand (AD)** can be split into two parts:

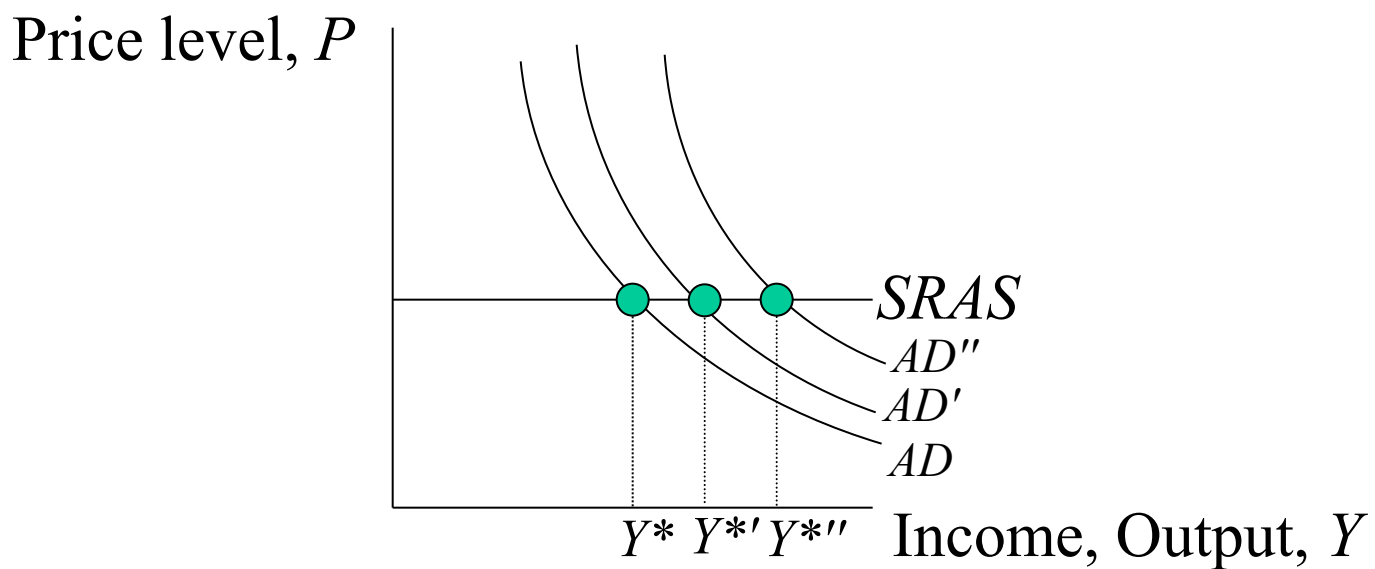
- **IS** model of the “**goods market**” and the
- **LM** model of the “**money market.**” “IS stands for Investment Saving, Whereas LM stands for Liquidity Money.”

The Keynesian model can be viewed as showing what causes the aggregate demand curve to shift.

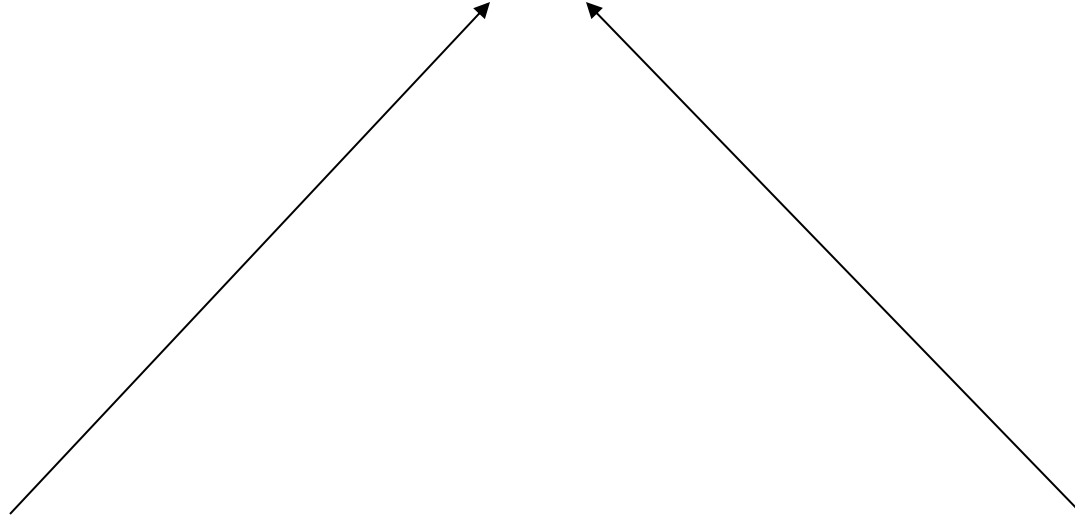


In the short run, when the price level is fixed, shifts in the aggregate demand curve lead to changes in national income, Y .

The model of aggregate demand developed in this chapter called the ***IS-LM*** is the leading interpretation of Keynes' work. **The *IS-LM* model takes the price level as given and shows what causes income to change. It shows what causes *AD* to shift.**



IS-LM Model of Aggregate Demand



IS (investment and saving)

model of the
'goods market'



LM (liquidity and money)
model of the 'money market'





The Goods Market and the IS Curve

The *IS curve* (which stands for investment saving) plots the relationship between the interest rate and the level of income that arises in the market for goods and services.



The Money Market and the LM Curve

The *LM curve* (which stands for liquidity and money) plots the relationship between the interest rate and the level of income that arises in the money market.



Because the **interest rate** influences both investment and money demand, it is the variable that links the two parts of the *IS-LM* model.

The model shows how interactions between these markets determine the position and slope of the aggregate demand curve, and therefore, the level of national income **in the short run**.

In the *General Theory of Money, Interest and Employment* (1936), Keynes proposed that an economy's total income was, in the short run, determined largely by the desire to spend by households, firms, and the government.

The more people want to spend, the more goods and services firms can sell. The more firms can sell, the more output they will choose to produce and the more workers they will choose to hire. Thus, the problem during recessions and depressions, according to Keynes, was inadequate spending.

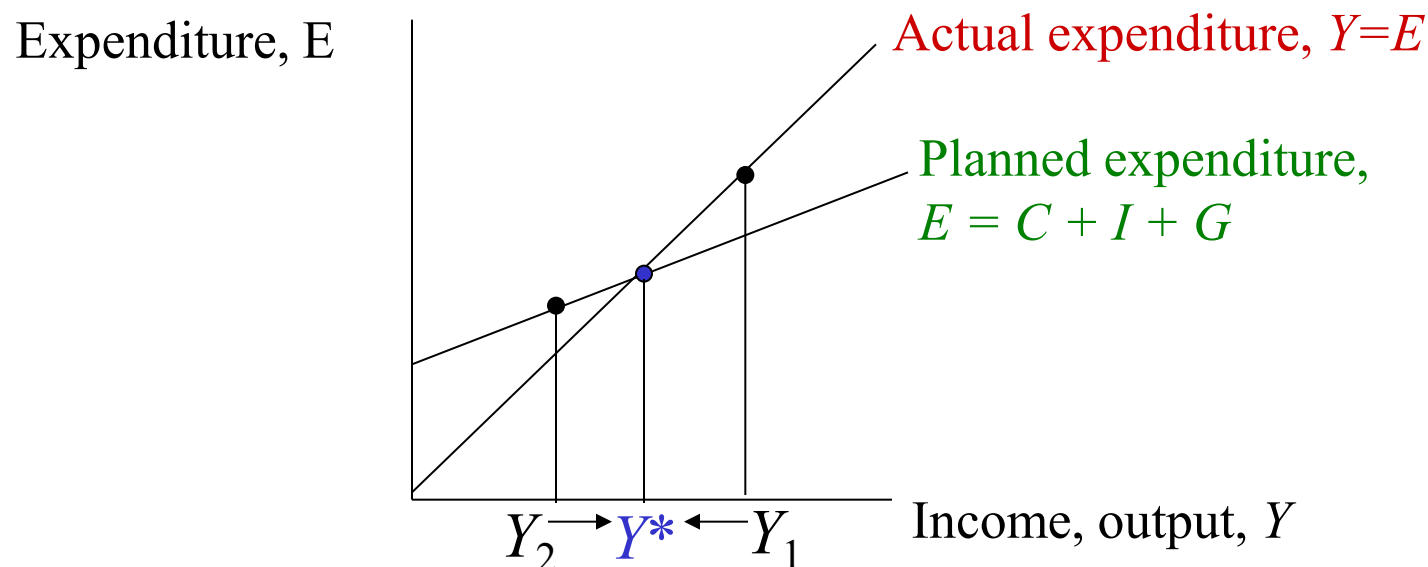
The **Keynesian cross** is an attempt to model this insight.

The Keynesian Cross

The Keynesian cross shows how income Y is determined for given levels of planned investment I and fiscal policy G and T . We can use this model to show how income changes when one of the exogenous variables change.

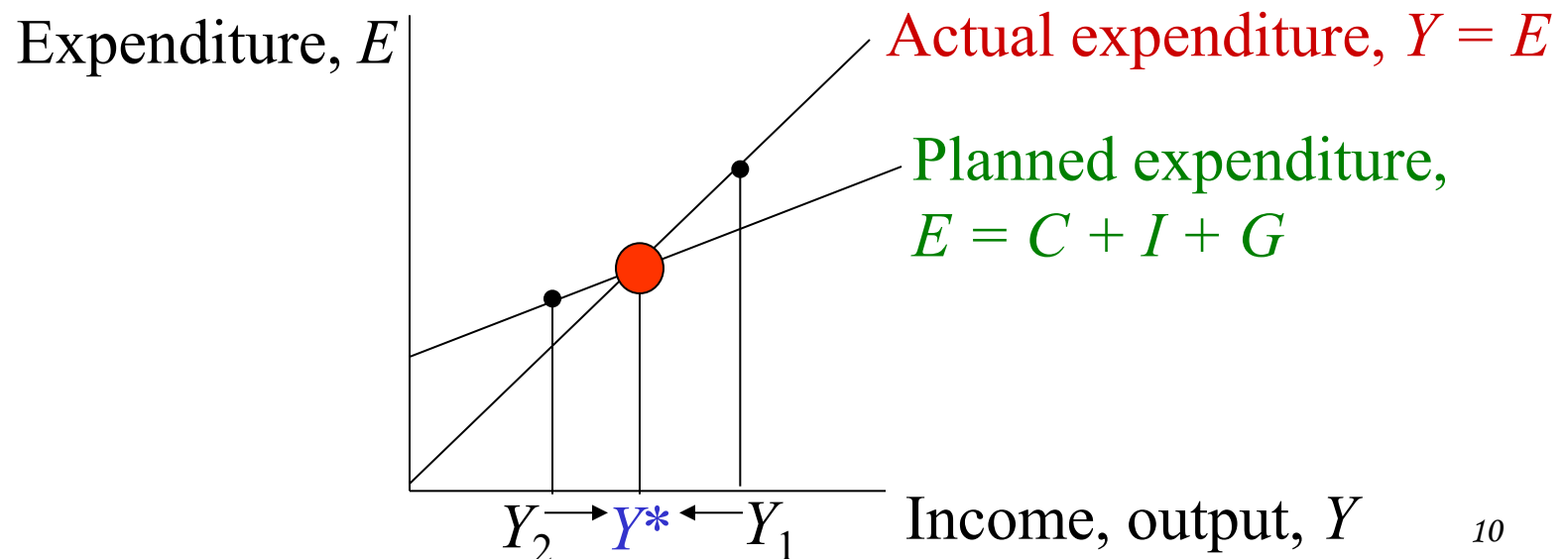
Actual expenditure is the amount households, firms and the government spend on goods and services (GDP).

Planned expenditure is the amount households, firms, and the government would like to spend on goods and services. The economy is in equilibrium when: **Actual Expenditure** = **Planned Expenditure** or $Y = E$.

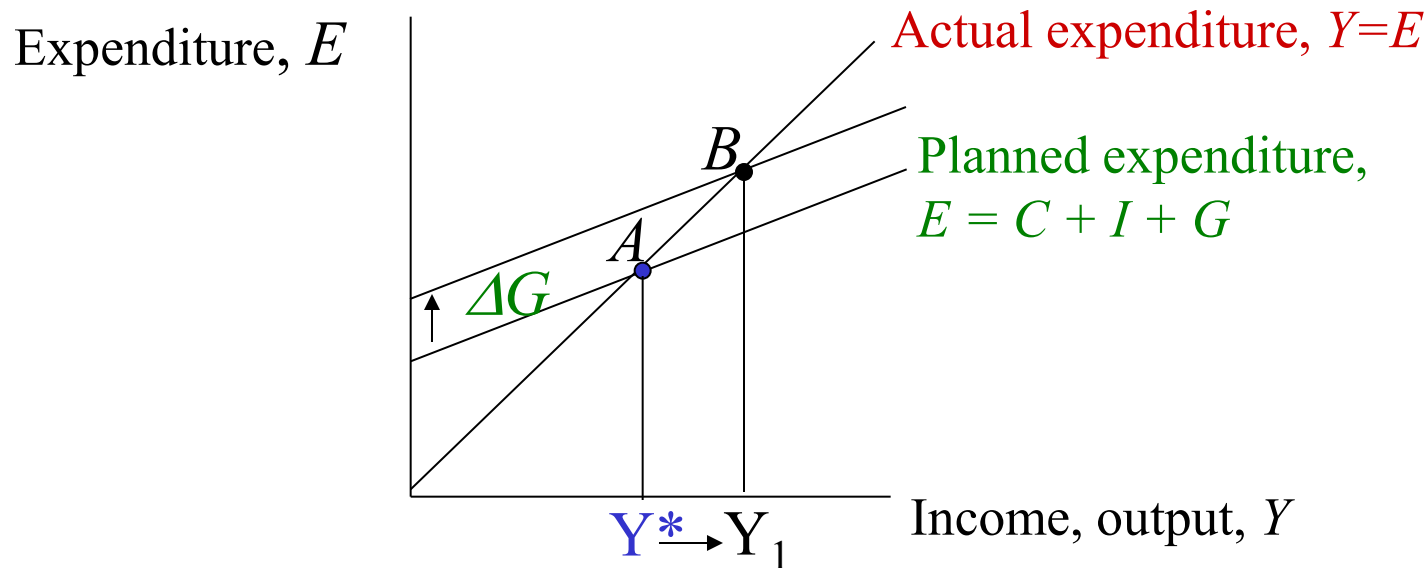


The 45-degree line ($Y=E$) plots the points where this condition holds. With the addition of the planned-expenditure function, this diagram becomes the **Keynesian cross**.

How does the economy get to this equilibrium? **Inventories play an important role in the adjustment process.** Whenever the economy is not in equilibrium, firms experience unplanned changes in inventories, and this induces them to change production levels. Changes in production in turn influence total income and expenditure, moving the economy toward equilibrium.



Consider how **changes in government purchases** affect the economy. Because government purchases are one component of expenditure, higher government purchases result in higher planned expenditure, for any given level of income.



An increase in government purchases of ΔG raises planned expenditure by that amount for any given level of income. The equilibrium moves from A to B and income rises. Note that the increase in income Y exceeds the increase in government purchases ΔG .

Thus, fiscal policy has a multiplied effect on income.

Fiscal Policy and the Multiplier

If government spending were to increase by \$1, then you might expect equilibrium output (Y) to also rise by \$1.

But it doesn't! The **multiplier shows** that the change in demand for output (Y) will be larger than the initial change in spending.

Here's why:

When there is an increase in government spending (ΔG), income rises by ΔG as well. The increase in income will raise consumption by $MPC \times \Delta G$, where MPC is the marginal propensity to consume.

The increase in consumption raises expenditure and income again. The second increase in income of $MPC \times \Delta G$ again raises consumption, this time by $MPC \times (MPC \times \Delta G)$, which again raises income and so on.

So, the multiplier process helps explain fluctuations in the demand for output. For example, if something in the economy decreases investment spending, then people whose incomes have decreased will spend less, thereby driving equilibrium demand down even further.

The government-purchases multiplier is:

$$\Delta Y / \Delta G = 1 + MPC + MPC^2 + MPC^3 + \dots$$

$$\Delta Y / \Delta G = 1 / 1 - MPC$$

The tax multiplier is:

$$\Delta Y / \Delta T = MPC / (1 - MPC)$$



A Mankiw
Macroeconomics
case study

Increasing Government Purchases to Stimulate the Economy: **The Obama Spending Plan**

When President Obama took office in 2009, the economy was undergoing a significant recession.

He proposed a package that would cost the government about **\$800 billion**, or about 5% of annual GDP. The package included some **tax cuts** and **higher transfer payments**, but much of it was made up of increases in **government purchases of goods and services**.

The Interest Rate, Investment and the IS Curve

Let's now add the relationship between the interest rate and investment to our model, writing the level of planned investment as: $I = I(r)$.

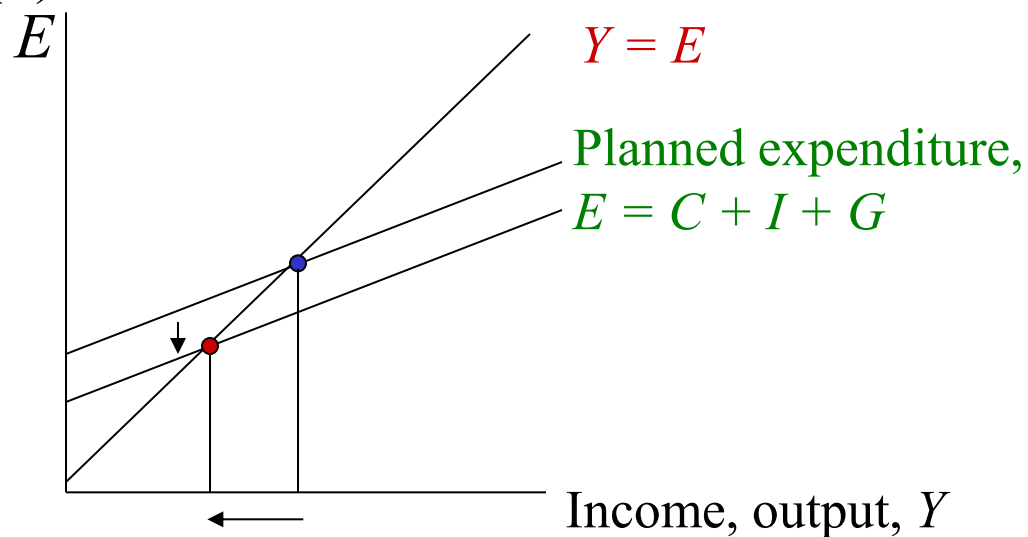
On the next slide, the investment function is graphed downward sloping showing the inverse relationship between investment and the interest rate. To determine how income changes when the interest rate changes, we combine the investment function with the Keynesian-cross diagram.

The *IS curve* summarizes this relationship between the interest rate and the level of income. In essence, the *IS* curve combines the interaction between I and Y demonstrated by the Keynesian cross. Because an increase in the interest rate causes planned investment to fall, which in turn causes income to fall, the *IS curve slopes downward*.

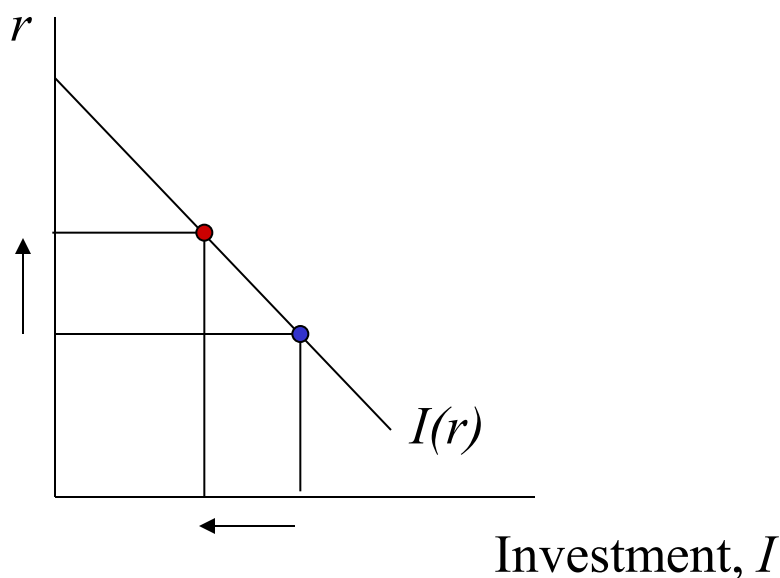
Deriving the IS Curve

An increase in the **interest rate** (in graph a), lowers planned investment, which shifts planned expenditure downward (in graph b) and lowers income (in graph c).

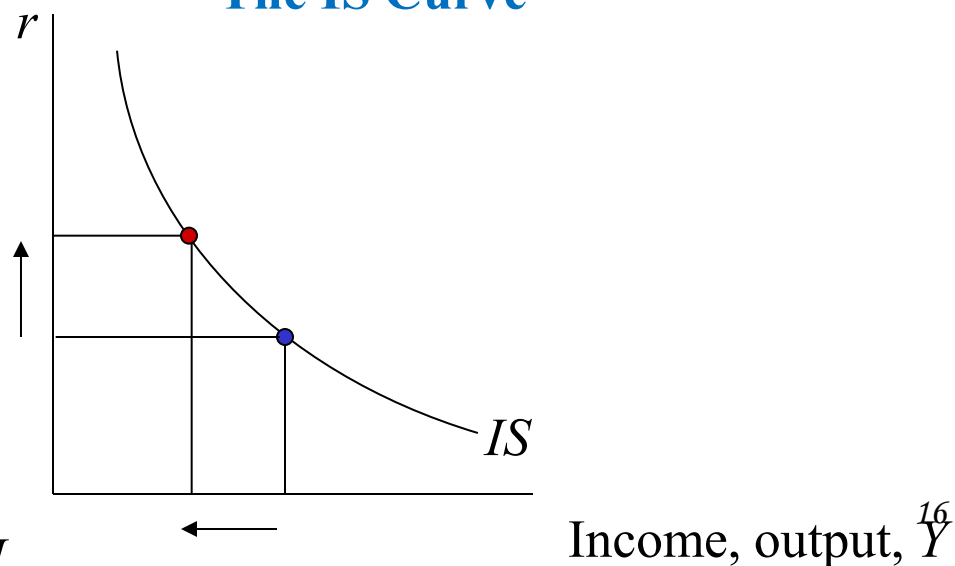
(b) **The Keynesian Cross**



(a) **The Investment Function**



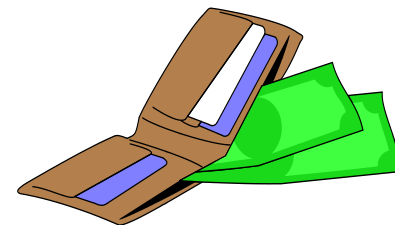
(c) **The IS Curve**



In summary, the ***IS* curve** shows the combinations of the **interest rate** and **the level of income** that are consistent with **equilibrium in the market for goods and services**.

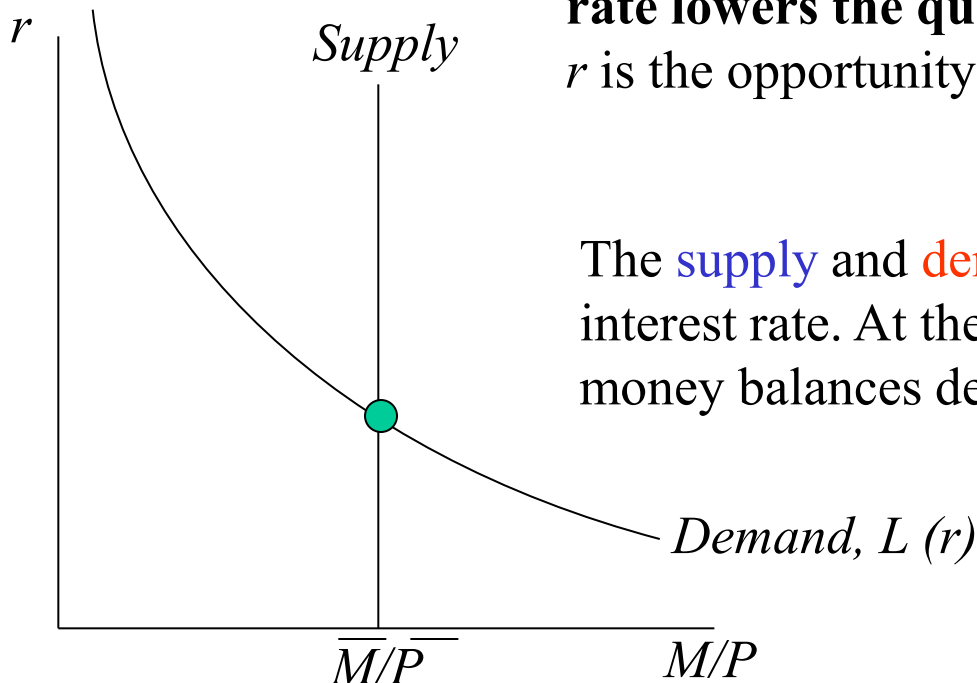
The *IS* curve is drawn for a given **fiscal policy**. Changes in fiscal policy that raise the demand for goods and services shift the *IS* curve to the right. Changes in fiscal policy that reduce the demand for goods and services shift the *IS* curve to the left.

The Money Market & LM Curve



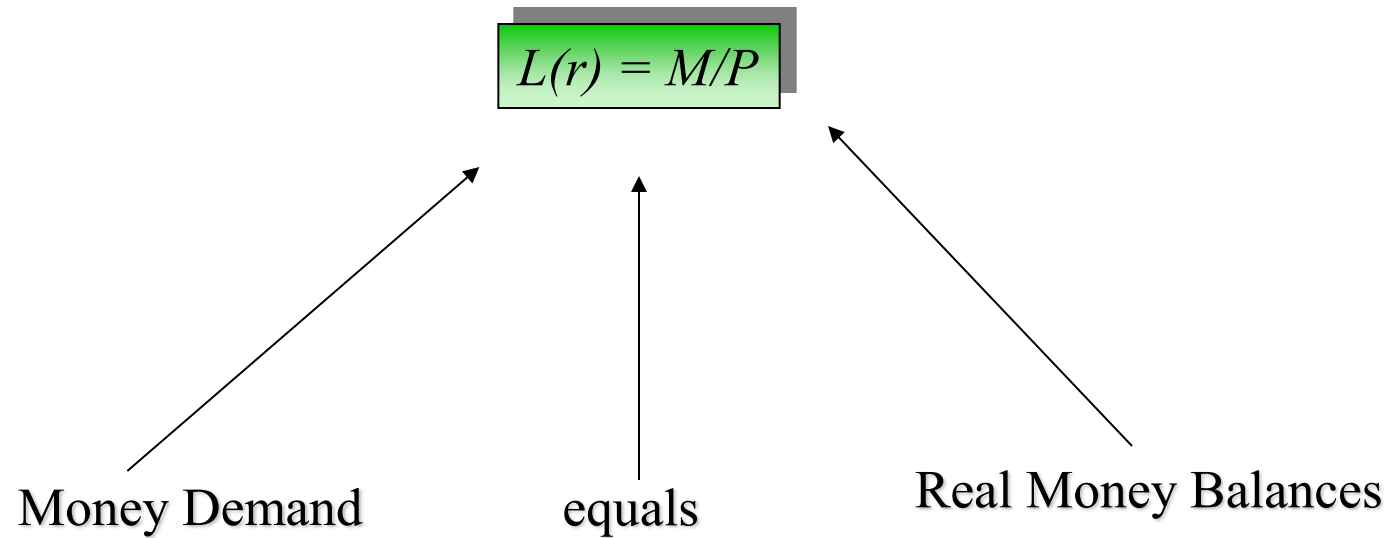
Now that we've derived the **IS part of AD**, it's now time to complete the model of AD by adding a money market equilibrium schedule, **the LM curve**. To develop this theory, we begin with the supply of real money balances (M/P); both of these variables are taken to be exogenously given. This yields a vertical supply curve.

Now, consider the demand for real money balances, L . The **theory of liquidity preference** suggests that a **higher interest rate lowers the quantity of real balances demanded**, because r is the opportunity cost of holding money.



The **supply** and **demand** for real money balances determine the interest rate. At the **equilibrium interest rate**, the quantity of money balances demanded equals the quantity supplied.

Money Market Equilibrium

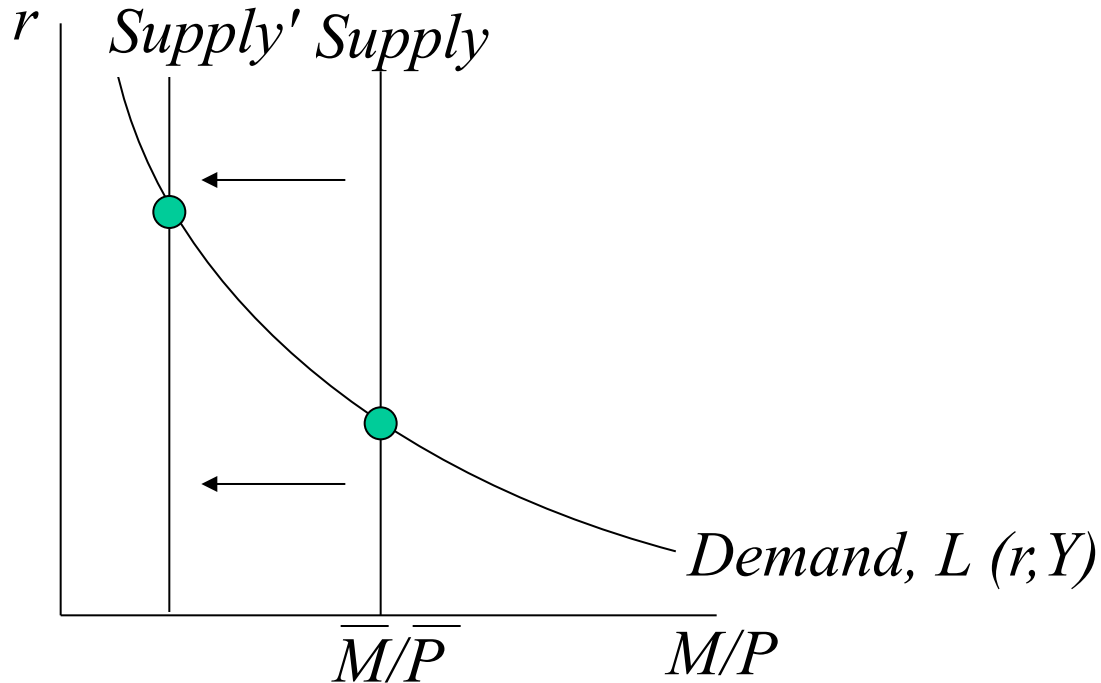


Money Demand

$$(M/P)^d = L(r, Y)$$

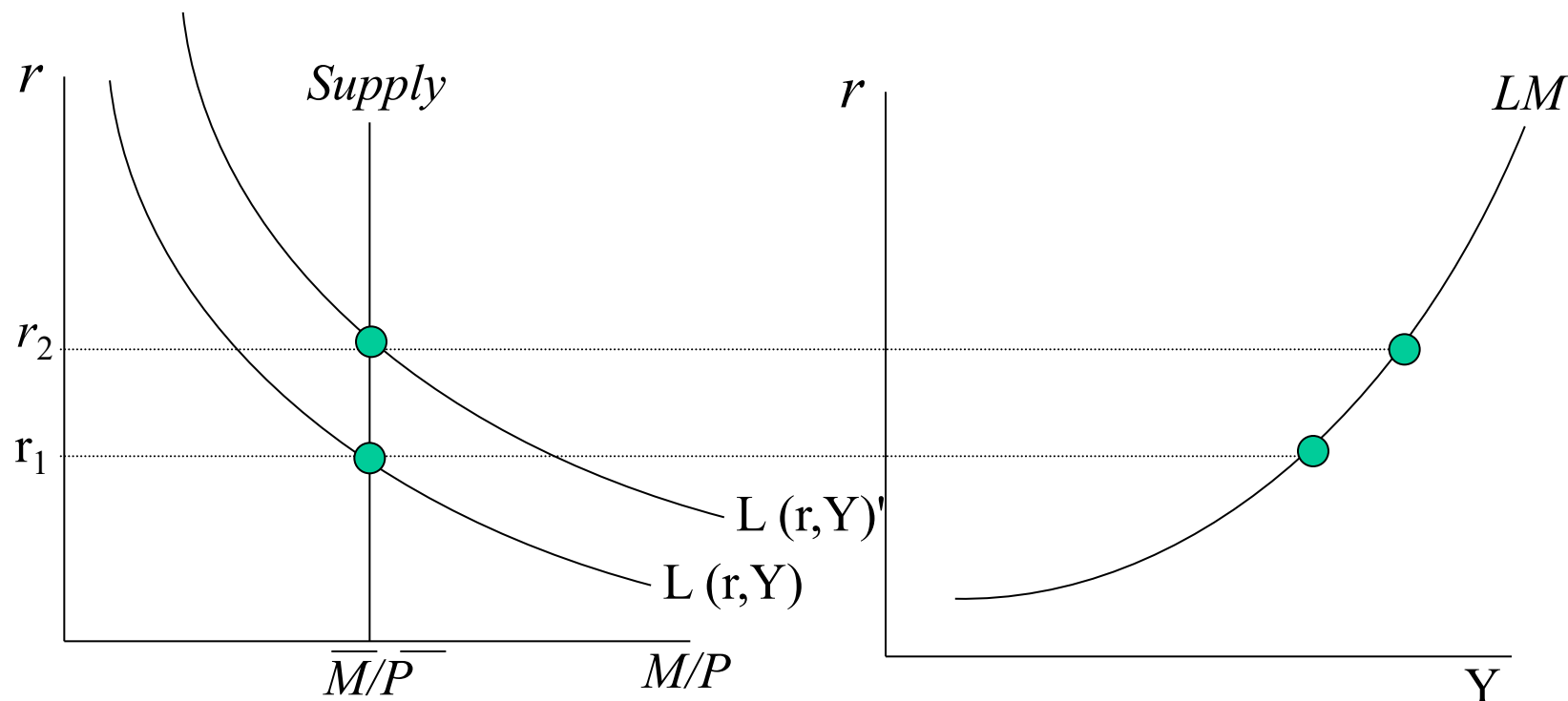
The **quantity of real money balances demanded** is negatively related to the **interest rate** (because r is the opportunity cost of holding money) and positively related to **income** (because of transactions demand).

A Reduction in the Money Supply: $-\Delta M/P$



Since the price level is fixed, a **reduction in the money supply** reduces the **supply of real balances**. Notice the **equilibrium interest rate rose**.

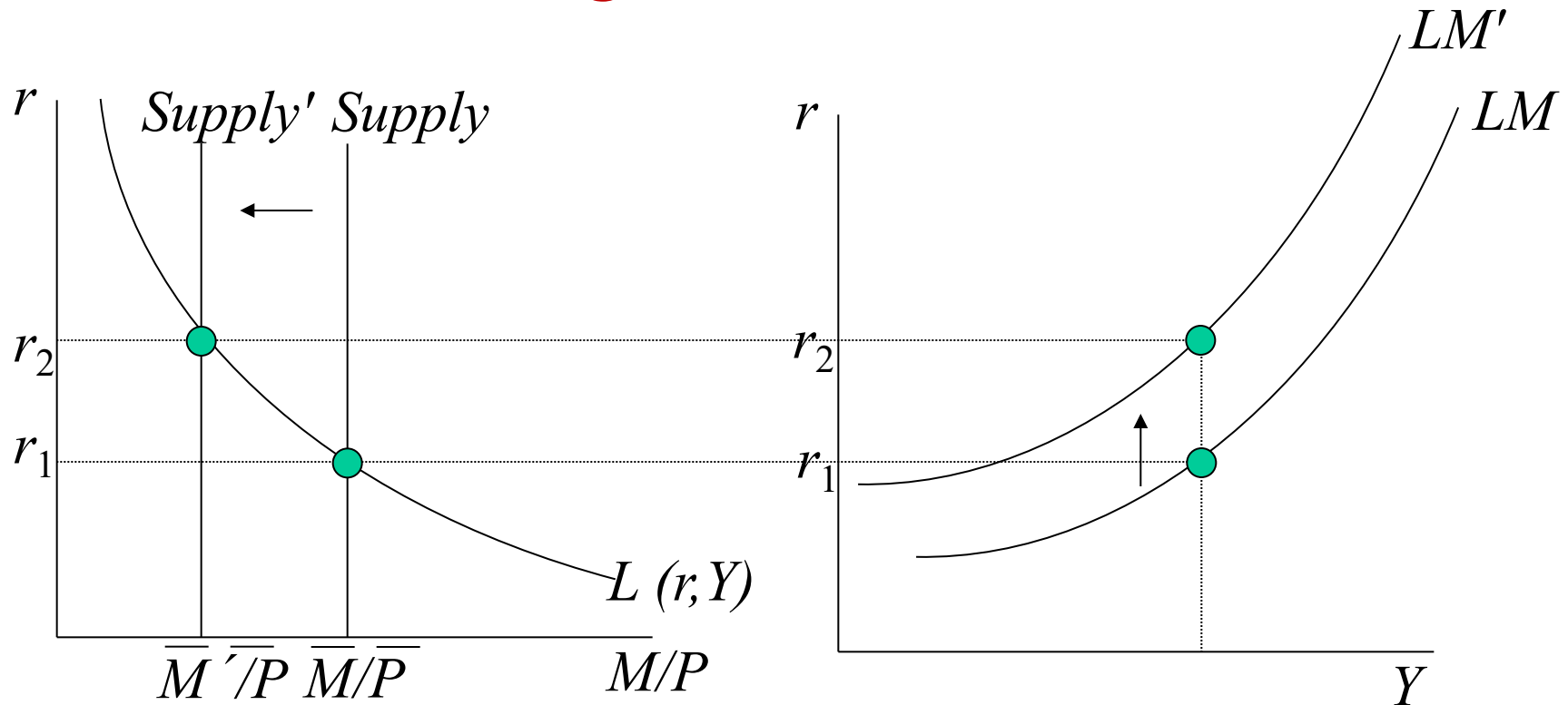
Deriving the LM Curve



An increase in income raises money demand, which increases the interest rate; this is called an increase in *transactions demand for money*.

The **LM curve** summarizes these changes in the money market equilibrium.

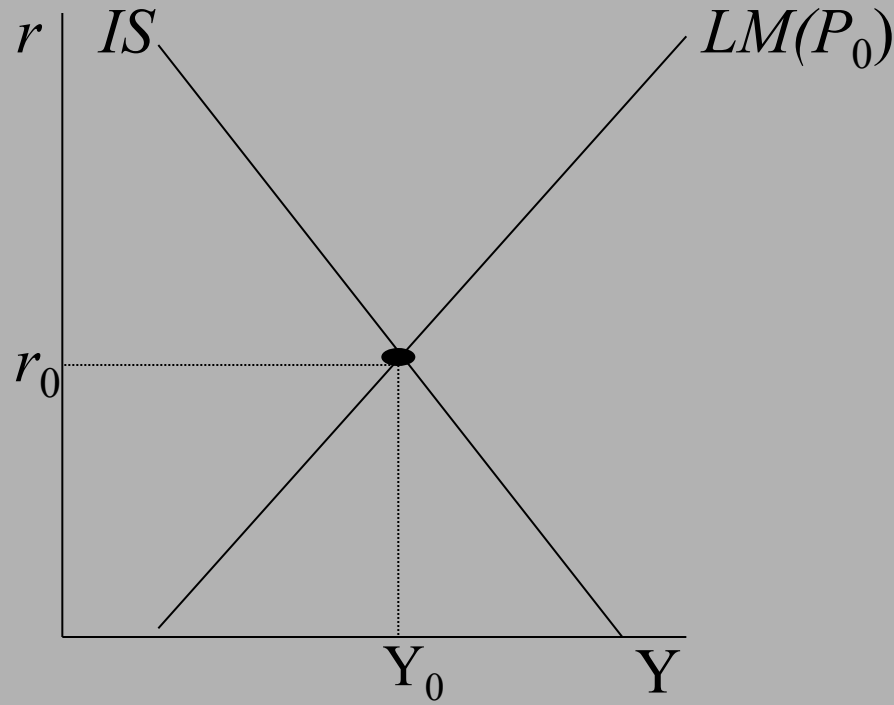
Shifting the LM Curve



A **contraction in the money supply** raises the interest rate that equilibrates the money market. Why?

Because a higher interest rate is needed to convince people to hold a smaller quantity of real balances. As a result of the decrease in the money supply, **LM shifts upward.**

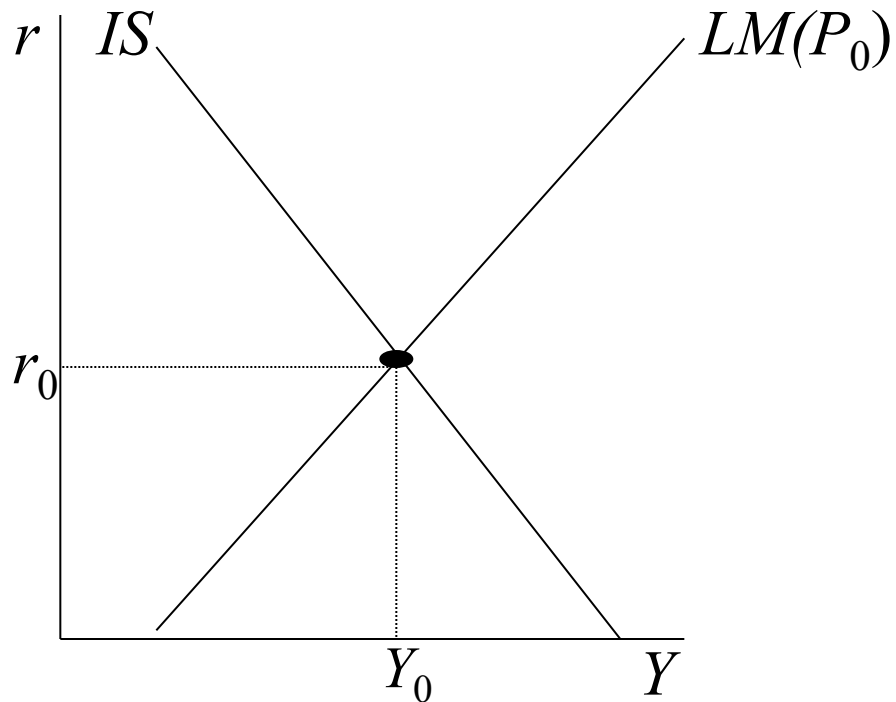
The IS-LM Model of AD



The intersection of the IS curve/equation, $Y = C(Y-T) + I(r) + G$ and the LM curve/equation $M/P = L(r, Y)$ determines the level of **aggregate demand**.

The intersection of the IS and LM curves represents **simultaneous equilibrium in the market for goods and services and in the market for real money balances** for given values of government spending, taxes, the money supply, and the price level.

The IS-LM Model of AD



Now that we've assembled the *IS-LM* model of aggregate demand, let's apply it to three issues:

- 1) Causes of fluctuations in national income
- 2) How *IS-LM* fits into the model of aggregate supply and aggregate demand in Chapter 9
- 3) The Great Depression

Explaining Fluctuations with the IS-LM Model

The intersection of the ***IS* curve and the *LM* curve** determines the level of **national income**, and the **interest rate** for a given price level.

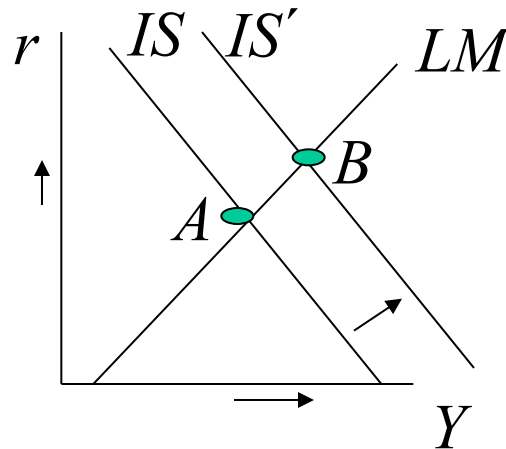
If the *IS* or *LM* curve shifts, the short-run equilibrium of the economy changes, and national income fluctuates.

Let's examine how changes in policy and shocks to the economy can cause these curves to shift.

How Fiscal Policy Shifts the IS Curve and Changes the Short-run Equilibrium?

$+\Delta G$

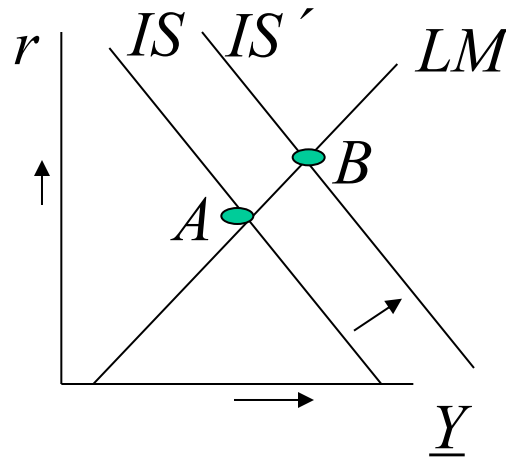
Consider an increase in government purchases.
This will raise the level of income by $\Delta G/(1 - MPC)$.



The IS curve shifts to the right by $\Delta G/(1 - MPC)$ which raises income and the interest rate.

$-\Delta T$

Consider a decrease in taxes of ΔT .
This will raise the level of income by $\Delta T \times MPC/(1 - MPC)$.

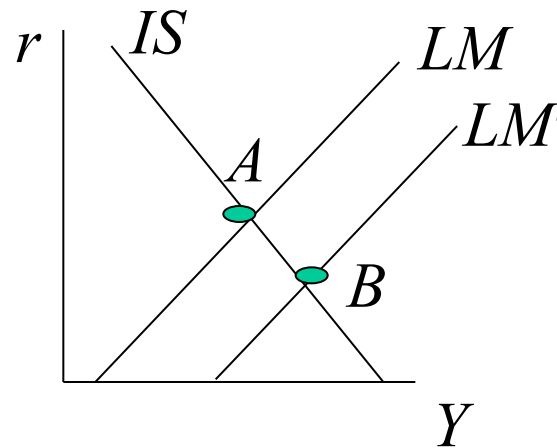


The IS curve shifts to the right by $\Delta T \times MPC/(1 - MPC)$ which raises income and the interest rate.

How Monetary Policy Shifts the LM Curve and Changes the Short-run Equilibrium?

$+\Delta M$

Consider an increase in the money supply.



The LM curve shifts downward and lowers the interest rate which raises income. Why?

Because when the Fed increases the supply of money, people have more money than they want to hold at the prevailing interest rate. As a result, they start depositing this extra money in banks or use it to buy bonds.

The interest rate r then falls until people are willing to hold all the extra money that the Fed has created; this brings the money market to a new equilibrium. The lower interest rate, in turn, has ramifications for the goods market. A lower interest rate stimulates planned investment, which increases planned expenditure, production, and income Y .

The *IS-LM* model shows that monetary policy influences income by changing the interest rate.

This conclusion sheds light on our analysis of monetary policy in Chapter 9. In that chapter we showed that in the short run, when prices are sticky, an expansion in the money supply raises income. But we didn't discuss how a monetary expansion induces greater spending on goods and services—a process called the *monetary transmission mechanism*.

The *IS-LM* model shows that an increase in the money supply lowers the interest rate, which stimulates investment and thereby expands the demand for goods and services.

Policy Analysis with Macroeconometric Models



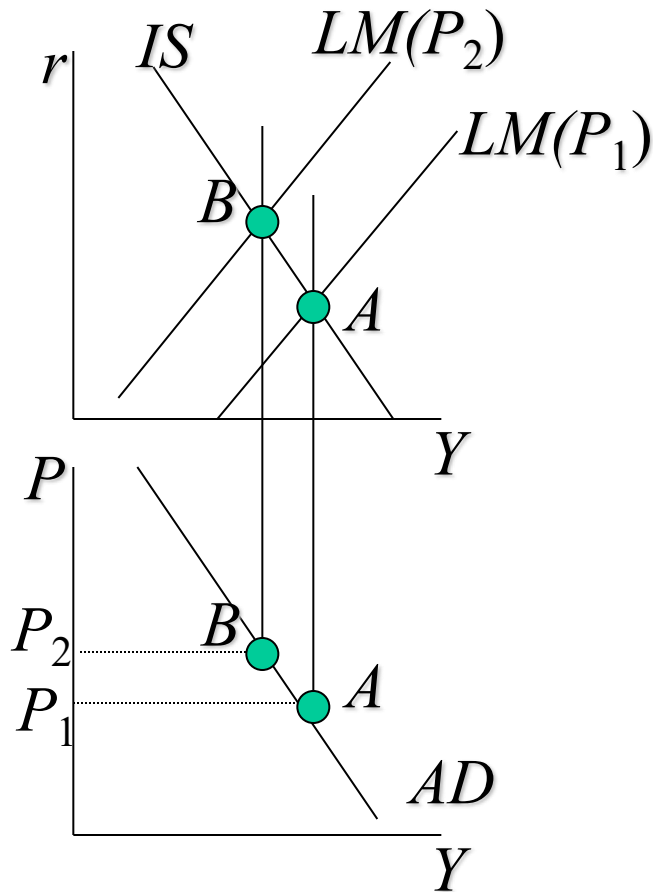
The IS-LM model shows how monetary and fiscal policy influence the equilibrium level of income. The predictions of the model, however, are qualitative, not quantitative. The *IS-LM* model that shows that **increases in government purchases raise GDP** and that **increases in taxes lower GDP**.

But, when economists analyze specific policy proposals, they must know the direction and size of the effect. Macroeconometric models describe the economy quantitatively, rather than just qualitatively.

IS-LM as a Theory of Aggregate Demand

From IS-LM to AD

You probably noticed from the *IS* and *LM* diagrams that r and Y were on the two axes. Now we're going to bring a third variable, the price level (P) into the analysis. We can accomplish this by linking both two-dimensional graphs.



To derive *AD*, start at point *A* in the top graph. Now increase the price level from P_1 to P_2 .

An increase in P lowers the value of real money balances, and Y , shifting *LM* leftward to point *B*.

Notice that r increased. Since r increased, we know that investment will decrease, as it just got more costly to take on various investment projects. This sets off a multiplier process since $-\Delta I$ causes $a - \Delta Y$. The $-\Delta Y$ triggers $-\Delta C$ as we move up the *IS* curve.

The $+\Delta P$ triggers a sequence of events that end with $a - \Delta Y$, the inverse relationship that defines the downward slope of *AD*.